EXPERIMENT 1
IMPLEMENTATION OF CAESAR CIPHER

AIM

To implement the **Caesar Cipher** substitution technique using C.

ALGORITHM

1. Read the plaintext from the user.
2. Read the key (shift value) from the user.
3. For each character in the plaintext:
    - If it is an uppercase letter, shift it by the key within A-Z.
    - If it is a lowercase letter, shift it by the key within a-z.
    - Leave non-alphabetic characters unchanged.
4. Display the encrypted (cipher) text.
5. Decrypt the cipher text by shifting each character backward using the same key.
6. Display the decrypted text.

CODE:

```
#include <stdio.h>
#include <string.h>

int main() {
    char text[100];
    int key, i;

    printf("Enter Plain Text: ");
    scanf("%s", text);

    printf("Enter Key: ");
    scanf("%d", &key);

    printf("Encrypted Text: ");
    for(i = 0; text[i] != '\0'; i++) {
        printf("%c", ((text[i]-'A'+key)%26)+'A');
    }

    printf("\nDecrypted Text: ");
    for(i = 0; text[i] != '\0'; i++) {
        printf("%c", ((text[i]-'A'+26-key)%26)+'A');
    }
```

```
    return 0;
}
```

```
Output                                                    Clear
Enter Plain Text: hello
Enter Key: 2
Encrypted Text: PMTTW
Decrypted Text: LIPPS

=== Code Execution Successful ===
```

RESULT

Thus, the **Caesar Cipher substitution technique** was implemented successfully using C.